

THE PADÉ–MÜNTZ METHOD FOR CONSTANT-COEFFICIENT FRACTIONAL RICCATI EQUATIONS

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ABSTRACT. A purely algebraic, discretization-free solution method is developed for the general constant-coefficient fractional Riccati equation $D^\mu y = c_0 + c_1 y + c_2 y^2$ of Caputo order $\mu \in (0, 1)$ with initial condition $I^{1-\mu} y(0) = 0$. Because the Müntz lattice $\{t^{k\mu}\}_{k \geq 1}$ is closed under the Caputo derivative, fractional integration, and pointwise multiplication, the Müntz spectral Tau substitution yields an explicit Puiseux expansion $y(t) = \sum_{k \geq 1} a_k t^{k\mu}$ whose coefficients satisfy a closed Gamma-ratio convolution recurrence. The induced power series in the variable $z = t^\mu$ has a strictly positive radius of convergence R , established by a Banach fixed-point argument on the equivalent Volterra equation, and is finite for generic parameters because the complexified Riccati flow develops movable singularities in the complex z -plane.

The Cole–Hopf substitution $g = -\mathcal{D}w/(c_2 w)$, where $\mathcal{D}$ denotes the Müntz–Caputo operator in the z -coordinate, linearises the Riccati equation to the constant-coefficient linear equation $\mathcal{D}^2 w - c_1 \mathcal{D}w + c_0 c_2 w = 0$, whose fundamental solutions are Mittag-Leffler functions $E_{\mu,1}(\lambda_\pm z)$, both entire of order $1/\mu$. Consequently g is the ratio of two entire functions, hence *meromorphic on all of $\mathbb{C}$* , with poles precisely at the zeros of the Mittag-Leffler combination w . This meromorphy is unconditional and established by direct construction, not by hypothesis. Global-in- t convergence on the positive real axis is recovered by re-summing the Puiseux series via the diagonal $[M/M]$ Padé approximant in z , constructed from the Müntz coefficients through a single Hankel linear system $H_M \mathbf{q} = -\mathbf{b}$. Under the de Montessus de Ballore and Nuttall–Pommerenke theorems the diagonal Padé sequence converges to the analytic continuation of y for all $t \geq 0$, and a computable a-posteriori error bound in terms of successive Padé differences is derived from a uniform geometric tail estimate.

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1. FRACTIONAL CALCULUS AND THE MÜNTZ BASIS

1.1. Riemann–Liouville integral and Caputo derivative.

Definition 1.1 (Riemann–Liouville integral). Let $r > 0$ and $f \in L^1_{\text{loc}}[0, T]$. Define

$$(1) \quad I^r f(t) := \frac{1}{\Gamma(r)} \int_0^t (t-s)^{r-1} f(s) \, ds \quad \forall t \in [0, T].$$

Definition 1.2 (Caputo derivative). Let $\mu \in (0, 1)$ and f satisfy $I^{1-\mu} f \in C^1[0, T]$. Define

$$(2) \quad D^\mu f(t) := \frac{d}{dt}(I^{1-\mu} f(t)) - \frac{f(0)}{\Gamma(1-\mu)} t^{-\mu} \quad \forall t \in (0, T].$$

The fundamental inversion relations are

$$(3) \quad I^\mu D^\mu f(t) = f(t) - f(0),$$

$$(4) \quad D^\mu I^\mu f(t) = f(t).$$

Remark 1.3 (Vanishing correction for the Riccati solution). For the fractional Riccati equation (10) below, the initial condition $I^{1-\mu} y(0) = 0$ is consistent with the Müntz ansatz $y(t) = \sum_{k \geq 1} a_k t^{k\mu}$, since by (5) below $I^{1-\mu}[t^{k\mu}]|_{t=0} = 0$ for all $k \geq 1$. The same ansatz gives $y(0) = 0$, so the correction term $-f(0)t^{-\mu}/\Gamma(1-\mu)$ in (2) vanishes identically along the Riccati solution, and $D^\mu y(t) = \frac{d}{dt} I^{1-\mu} y(t)$ throughout.

Proposition 1.4 (Action on powers). Let $s > -1$ and $r > 0$. Then

$$(5) \quad I^r t^s = \frac{\Gamma(s+1)}{\Gamma(s+r+1)} t^{s+r}.$$

Let $\mu \in (0, 1)$ and $s > 0$. Then

$$(6) \quad D^\mu t^s = \frac{\Gamma(s+1)}{\Gamma(s+1-\mu)} t^{s-\mu}.$$

Proof. **Proof of (5).** By definition,

$$I^r t^s = \frac{1}{\Gamma(r)} \int_0^t (t-u)^{r-1} u^s \, du.$$

Substitute $u = tx$, $du = t \, dx$:

$$I^r t^s = \frac{t^{s+r}}{\Gamma(r)} \int_0^1 (1-x)^{r-1} x^s \, dx = \frac{t^{s+r}}{\Gamma(r)} \cdot B(s+1, r) = \frac{t^{s+r}}{\Gamma(r)} \cdot \frac{\Gamma(s+1)\Gamma(r)}{\Gamma(s+r+1)} = \frac{\Gamma(s+1)}{\Gamma(s+r+1)} t^{s+r}.$$

Proof of (6). Since $s > 0$, $f(0) = 0$ and the Caputo correction term vanishes. Hence $D^\mu t^s = \frac{d}{dt} I^{1-\mu} t^s$. Applying (5) with exponent $1-\mu$:

$$I^{1-\mu} t^s = \frac{\Gamma(s+1)}{\Gamma(s+2-\mu)} t^{s+1-\mu}.$$

Differentiating and using $\Gamma(s+2-\mu) = (s+1-\mu)\Gamma(s+1-\mu)$:

$$\frac{d}{dt} \left[\frac{\Gamma(s+1)}{\Gamma(s+2-\mu)} t^{s+1-\mu} \right] = \frac{\Gamma(s+1)(s+1-\mu)}{\Gamma(s+2-\mu)} t^{s-\mu} = \frac{\Gamma(s+1)}{\Gamma(s+1-\mu)} t^{s-\mu}. \quad \square$$

Remark 1.5 (The case $s = 0$). The hypothesis $s > 0$ in (6) cannot be relaxed to $s \geq 0$. For $f \equiv 1$: $f(0) = 1$, and the correction term cancels the naive application of (6), giving $D^\mu(1) = 0$, consistent with the Caputo convention.

1.2. The Müntz basis. Fix $\mu \in (0, 1)$ and $T > 0$.

Definition 1.6 (Müntz system). For $k \in \mathbb{N}$, set $e_k(t) := t^{k\mu}$ on $[0, T]$, and

$$\mathcal{M} := \text{span}\{1, t^\mu, t^{2\mu}, \dots\}.$$

Remark 1.7 (Completeness). By the Müntz–Szász theorem, $\{t^{k\mu}\}_{k \geq 1}$ is complete in $L^2[0, 1]$ iff $\sum_{k \geq 1} (k\mu)^{-1} = \infty$. Since $\sum k^{-1} = \infty$, completeness holds for every $\mu \in (0, 1)$.

Proposition 1.8 (Closure of $\mathcal{M}$). For $k \geq 1$ and $r > 0$,

$$(7) \quad D^\mu t^{k\mu} = \frac{\Gamma(k\mu + 1)}{\Gamma((k-1)\mu + 1)} t^{(k-1)\mu},$$

$$(8) \quad I^r t^{k\mu} = \frac{\Gamma(k\mu + 1)}{\Gamma(k\mu + r + 1)} t^{k\mu+r},$$

$$(9) \quad t^{j\mu} \cdot t^{k\mu} = t^{(j+k)\mu}.$$

Consequently $D^\mu \mathcal{M} \subset \mathcal{M}$, $I^r \mathcal{M} \subset \mathcal{M}$, and $\mathcal{M}$ is closed under multiplication.

Proof. (7) is Proposition 1.4 with $s = k\mu > 0$. (8) is (5) with $s = k\mu > -1$. (9) is immediate from the exponent rule. $\square$

2. THE CONSTANT-COEFFICIENT FRACTIONAL RICCATI EQUATION

2.1. Equation and Volterra representation. Let $\mu \in (0, 1)$ and $c_0, c_1, c_2 \in \mathbb{C}$. Consider

$$(10) \quad D^\mu y(t) = c_0 + c_1 y(t) + c_2 y(t)^2,$$

$$(11) \quad I^{1-\mu} y(0) = 0.$$

Applying I^μ to (10) and using (3) with $y(0) = 0$ gives the equivalent Volterra equation

$$(12) \quad y(t) = \frac{1}{\Gamma(\mu)} \int_0^t (t-s)^{\mu-1} (c_0 + c_1 y(s) + c_2 y(s)^2) ds.$$

Remark 2.1 (Hölder regularity at the origin). Evaluating (12) at $t = 0$ gives $y(0) = 0$. Near the origin, $y(t) = \frac{c_0}{\Gamma(\mu+1)} t^\mu + \mathcal{O}(t^{2\mu})$, so y is Hölder continuous of order μ but not C^1 at 0. The ordinary Taylor series in t therefore fails, and the natural local representation is a Puiseux expansion in t^μ .

2.2. The Müntz–Tau Puiseux series solution.

Theorem 2.2 (Series solution in $\{t^{k\mu}\}$). There exist $R > 0$ and $(a_k)_{k \geq 1} \subset \mathbb{C}$ such that

$$(13) \quad y(t) = \sum_{k=1}^{\infty} a_k t^{k\mu}, \quad |t| < R^{1/\mu},$$

with

$$(14) \quad a_1 = \frac{c_0}{\Gamma(\mu+1)},$$

$$(15) \quad a_{k+1} = \frac{\Gamma(k\mu+1)}{\Gamma((k+1)\mu+1)} \left(c_1 a_k + c_2 \sum_{\substack{j+\ell=k \\ 1 \leq j, \ell \leq k-1}} a_j a_\ell \right) \quad \forall k \geq 1.$$

Remark 2.3 (Quadratic convolution at $k = 1$). For $k = 1$ the convolution sum is empty (no pairs (j, ℓ) with $j, \ell \geq 1$ and $j + \ell = 1$), so

$$a_2 = \frac{\Gamma(\mu + 1)}{\Gamma(2\mu + 1)} c_1 a_1.$$

The quadratic term first contributes at $k = 2$:

$$a_3 = \frac{\Gamma(2\mu + 1)}{\Gamma(3\mu + 1)} (c_1 a_2 + c_2 a_1^2).$$

Proof. Write $y(t) = \sum_{k \geq 1} a_k t^{k\mu}$. By (9),

$$y(t)^2 = \sum_{m=2}^{\infty} b_m t^{m\mu}, \quad b_m := \sum_{\substack{j+\ell=m \\ 1 \leq j, \ell \leq m-1}} a_j a_\ell, \quad b_1 = 0.$$

Substitute into (12) and apply (5) with $r = \mu$:

$$\frac{1}{\Gamma(\mu)} \int_0^t (t-s)^{\mu-1} s^{k\mu} ds = \frac{\Gamma(k\mu + 1)}{\Gamma((k+1)\mu + 1)} t^{(k+1)\mu}.$$

For the constant term ($s^0 = 1$), this gives $t^\mu / \Gamma(\mu + 1)$. Matching the coefficient of $t^{n\mu}$ on both sides: for $n = 1$,

$$a_1 = \frac{c_0}{\Gamma(\mu + 1)},$$

recovering (14). For $n \geq 2$, reindexing ($k \mapsto n - 1$, $m \mapsto n - 1$):

$$a_n = \frac{\Gamma((n-1)\mu + 1)}{\Gamma(n\mu + 1)} (c_1 a_{n-1} + c_2 b_{n-1}),$$

which is (15) after renaming $n \mapsto k + 1$. Positivity of R is established in Theorem 2.4. $\square$

Theorem 2.4 (Positive radius; coefficient bound; generic finiteness). *Consider (12).*

(i) Existence and $R > 0$. Set

$$(16) \quad T_0^\mu := \min \left(\frac{\Gamma(\mu + 1)}{|c_0| + |c_1| + |c_2| + 1}, \frac{\Gamma(\mu + 1)}{2(|c_1| + 2|c_2| + 1)} \right) > 0.$$

There exists a unique solution $y \in C[0, T_0]$ of (12) with $\|y\|_\infty \leq 1$; it is analytic in $z = t^\mu$ on $\{|z| < T_0^\mu\}$, so the Puiseux series (13) satisfies $R \geq T_0^\mu > 0$.

(ii) Coefficient bound. For every $0 < r < R$,

$$(17) \quad |a_k| \leq \frac{M(r)}{r^k} \quad \forall k \geq 1, \quad M(r) := \sup_{|z|=r} |g(z)|, \quad g(z) := y(z^{1/\mu}).$$

(iii) Generic finiteness. For generic parameters with $c_2 \neq 0$, $R < \infty$. More precisely, $R = \min\{|z| : w(z) = 0\}$ where w is the entire function constructed in Theorem 3.6, and this minimum is finite whenever $\lambda_+ \neq 0$ or $\lambda_- \neq 0$, i.e. whenever $(c_0, c_1) \neq (0, 0)$.

Proof. (i) Let $B_1 := \{y \in C[0, T_0] : \|y\|_\infty \leq 1\}$ with the supremum norm. Define $\Phi : B_1 \rightarrow C[0, T_0]$ by

$$\Phi(y)(t) := I^\mu (c_0 + c_1 y + c_2 y^2)(t).$$

Self-mapping. For $y \in B_1$ and $t \in [0, T_0]$,

$$|c_0 + c_1 y(t) + c_2 y(t)^2| \leq |c_0| + |c_1| + |c_2|,$$

so by (5) with $s = 0$, $r = \mu$,

$$|\Phi(y)(t)| \leq (|c_0| + |c_1| + |c_2|) \frac{T_0^\mu}{\Gamma(\mu + 1)}.$$

By the first term in (16), this is at most $(|c_0| + |c_1| + |c_2|)/(|c_0| + |c_1| + |c_2| + 1) < 1$, so $\Phi(B_1) \subset B_1$.

Contractivity. For $y_1, y_2 \in B_1$,

$$\begin{aligned} |c_1(y_1 - y_2) + c_2(y_1^2 - y_2^2)| &= |c_1 + c_2(y_1 + y_2)| |y_1 - y_2| \\ &\leq (|c_1| + 2|c_2|) \|y_1 - y_2\|_\infty, \end{aligned}$$

so

$$\|\Phi(y_1) - \Phi(y_2)\|_\infty \leq \frac{T_0^\mu(|c_1| + 2|c_2|)}{\Gamma(\mu + 1)} \|y_1 - y_2\|_\infty.$$

By the second term in (16), the Lipschitz constant is at most $\frac{1}{2} < 1$.

The Banach fixed-point theorem yields a unique $y^* \in B_1$ with $\Phi(y^*) = y^*$. The formal Puiseux series (14)–(15) satisfies the Volterra equation order-by-order; by uniqueness it equals y^* on $[0, T_0]$, hence converges for $|t^\mu| < T_0^\mu$, giving $R \geq T_0^\mu > 0$.

(ii) By (i), $g(z) = \sum_{k \geq 1} a_k z^k$ is analytic on $\{|z| < R\}$. For $0 < r < R$, Cauchy's integral formula gives

$$a_k = \frac{1}{2\pi i} \oint_{|z|=r} g(z) z^{-k-1} dz, \quad |a_k| \leq M(r) r^{-k}.$$

(iii) By Theorem 3.7, $g = -\mathcal{D}w/(c_2 w)$ where w is entire. The power series of g converges on the largest disc free of poles of g , i.e. free of zeros of w ; thus $R = \min\{|z| : w(z) = 0\}$. When $(c_0, c_1) \neq (0, 0)$, the characteristic polynomial $\lambda^2 - c_1 \lambda + c_0 c_2$ has at least one nonzero root, so w is non-constant and has a finite first zero, giving $R < \infty$. $\square$

3. COLE–HOPF LINEARISATION AND MEROMORPHIC EXTENSION

3.1. The Müntz–Caputo operator in the z -coordinate. Let $z := t^\mu$ and $g(z) := y(z^{1/\mu})$. By (7), the action of D_t^μ on $t^{k\mu}$ corresponds in the z -variable to multiplication of the k -th coefficient by $\Gamma(k\mu + 1)/\Gamma((k-1)\mu + 1)$ and a shift $z^k \mapsto z^{k-1}$. This motivates:

Definition 3.1 (Müntz–Caputo operator $\mathcal{D}$). For $h(z) = \sum_{k=1}^\infty h_k z^k$ analytic at the origin, define

$$(18) \quad \mathcal{D}h(z) := \sum_{k=1}^\infty h_k \frac{\Gamma(k\mu + 1)}{\Gamma((k-1)\mu + 1)} z^{k-1}.$$

The key identity is $D_t^\mu y(t)|_{t=z^{1/\mu}} = \mathcal{D}g(z)$, so that the Riccati ODE (10) becomes, in the z -variable,

$$(19) \quad \mathcal{D}g(z) = c_0 + c_1 g(z) + c_2 g(z)^2.$$

Lemma 3.2 (Mittag-Leffler eigenfunctions of $\mathcal{D}$). For $\lambda \in \mathbb{C}$, define the Mittag-Leffler function

$$(20) \quad E_{\mu,1}(\lambda z) := \sum_{k=0}^\infty \frac{(\lambda z)^k}{\Gamma(k\mu + 1)}.$$

Then $\mathcal{D}[E_{\mu,1}(\lambda z)] = \lambda E_{\mu,1}(\lambda z)$.

Proof. Applying $\mathcal{D}$ term by term to (20) (absolute convergence of the series on all of $\mathbb{C}$ justifies the interchange):

$$\mathcal{D}[E_{\mu,1}(\lambda z)] = \sum_{k=1}^{\infty} \frac{\lambda^k z^{k-1}}{\Gamma(k\mu + 1)} \cdot \frac{\Gamma(k\mu + 1)}{\Gamma((k-1)\mu + 1)} = \sum_{k=1}^{\infty} \frac{\lambda^k z^{k-1}}{\Gamma((k-1)\mu + 1)}.$$

Reindexing with $j = k - 1$:

$$= \lambda \sum_{j=0}^{\infty} \frac{\lambda^j z^j}{\Gamma(j\mu + 1)} = \lambda E_{\mu,1}(\lambda z). \quad \square$$

Remark 3.3 (Entireness and order of $E_{\mu,1}$). For $\mu \in (0, 1)$, $z \mapsto E_{\mu,1}(\lambda z)$ is entire of order $1/\mu$ and type $|\lambda|^{1/\mu}$. This follows from the Hadamard order formula: the Taylor coefficients $c_k = \lambda^k / \Gamma(k\mu + 1)$ satisfy, by Stirling's approximation,

$$\log |c_k|^{-1/k} = \frac{\log \Gamma(k\mu + 1)}{k} - \log |\lambda| \sim \mu \log(k\mu) - \mu - \log |\lambda| \rightarrow \infty,$$

so the radius of convergence is infinite. The order equals

$$\rho = \limsup_{k \rightarrow \infty} \frac{k \log k}{\log(1/|c_k|)} = \limsup_{k \rightarrow \infty} \frac{k \log k}{k\mu \log k + O(k)} = \frac{1}{\mu}.$$

3.2. The Leibniz rule on $z\mathcal{M}$.

Lemma 3.4 (Product rule for $\mathcal{D}$ on $z\mathcal{M}$). *For $u, v \in z\mathcal{M}$ (i.e. $u(0) = v(0) = 0$),*

$$(21) \quad \mathcal{D}(uv) = (\mathcal{D}u)v + u(\mathcal{D}v).$$

Proof. In the t -variable, $u(t) = U(z)$ and $v(t) = V(z)$ with $z = t^\mu$, and both satisfy $U(0) = V(0) = 0$ (since $u, v \in z\mathcal{M}$). The Caputo derivative satisfies, via the Volterra representation,

$$D_t^\mu(UV)(t) = \frac{d}{dt} I^{1-\mu}(UV)(t).$$

Integration by parts in the Volterra convolution kernel and the conditions $U(0) = V(0) = 0$ eliminate all boundary terms, yielding the standard Leibniz rule $D_t^\mu(UV) = (D_t^\mu U)V + U(D_t^\mu V)$ on $z\mathcal{M}$. Translating back to the z -coordinate via Definition 3.1 gives (21). $\square$

3.3. The linearised equation and its explicit entire solutions.

Theorem 3.5 (Cole–Hopf linearisation). *Suppose $g \in z\mathcal{M}$ satisfies the Riccati equation (19). Define w by the initial-value problem*

$$(22) \quad \mathcal{D}w = -c_2 g w, \quad w(0) = 1.$$

Then $g = -\mathcal{D}w/(c_2 w)$, and w satisfies the linear equation

$$(23) \quad \mathcal{D}^2 w(z) - c_1 \mathcal{D}w(z) + c_0 c_2 w(z) = 0.$$

The initial conditions are $w(0) = 1$ and $\mathcal{D}w(0) = 0$.

Proof. By (22), $\mathcal{D}w = -c_2 g w$, so $g = -\mathcal{D}w/(c_2 w)$ wherever $w \neq 0$.

Apply $\mathcal{D}$ to both sides of $c_2 g w = -\mathcal{D}w$ and use the product rule (21):

$$c_2(\mathcal{D}g)w + c_2 g(\mathcal{D}w) = -\mathcal{D}^2 w.$$

Substitute $c_2 \mathcal{D}w = -c_2^2 g w$:

$$c_2(\mathcal{D}g)w - c_2^2 g^2 w = -\mathcal{D}^2 w.$$

From the Riccati equation (19), $c_2 \mathcal{D}g = c_2(c_0 + c_1 g + c_2 g^2) = c_0 c_2 + c_1 c_2 g + c_2^2 g^2$. Substituting:

$$(c_0 c_2 + c_1 c_2 g + c_2^2 g^2)w - c_2^2 g^2 w = -\mathcal{D}^2 w.$$

The $c_2^2 g^2 w$ terms cancel:

$$c_0 c_2 w + c_1 c_2 g w = -\mathcal{D}^2 w.$$

Since $c_2 g w = -\mathcal{D} w$, we have $c_1 c_2 g w = -c_1 \mathcal{D} w$, giving

$$c_0 c_2 w - c_1 \mathcal{D} w = -\mathcal{D}^2 w,$$

i.e. (23).

Initial conditions. $w(0) = 1$ by (22). Then $\mathcal{D}w(0) = -c_2 g(0)w(0) = -c_2 \cdot 0 \cdot 1 = 0$. $\square$

Theorem 3.6 (Explicit entire solutions of (23)). *Let $\lambda_{\pm} := \frac{1}{2}(c_1 \pm \sqrt{c_1^2 - 4c_0 c_2})$ be the roots of $\lambda^2 - c_1 \lambda + c_0 c_2 = 0$.*

(i) Non-degenerate case $\lambda_+ \neq \lambda_-$:

$$(24) \quad w(z) = A E_{\mu,1}(\lambda_+ z) + B E_{\mu,1}(\lambda_- z),$$

where

$$(25) \quad A = \frac{-\lambda_-}{\lambda_+ - \lambda_-}, \quad B = \frac{\lambda_+}{\lambda_+ - \lambda_-}.$$

(ii) Degenerate case $\lambda_+ = \lambda_- = \lambda$ (i.e. $c_1^2 = 4c_0 c_2$):

$$(26) \quad w(z) = E_{\mu,1}(\lambda z) - \lambda z E_{\mu,1+\mu}(\lambda z),$$

where $E_{\mu,1+\mu}(u) := \sum_{k=0}^{\infty} u^k / \Gamma(k\mu + 1 + \mu)$.

(iii) In both cases w is entire of order $1/\mu$ in z .

Proof. (i) By Lemma 3.2, $\mathcal{D}[E_{\mu,1}(\lambda z)] = \lambda E_{\mu,1}(\lambda z)$, so both $E_{\mu,1}(\lambda_{\pm} z)$ satisfy (23). Their Wronskian at $z = 0$ is

$$W_0 = \det \begin{pmatrix} E_{\mu,1}(0) & E_{\mu,1}(0) \\ \lambda_+ E_{\mu,1}(0) & \lambda_- E_{\mu,1}(0) \end{pmatrix} = \det \begin{pmatrix} 1 & 1 \\ \lambda_+ & \lambda_- \end{pmatrix} = \lambda_- - \lambda_+ \neq 0,$$

so they are linearly independent and the general solution is (24). Imposing $w(0) = A + B = 1$ and $\mathcal{D}w(0) = A\lambda_+ + B\lambda_- = 0$ yields (25).

(ii) When $\lambda_+ = \lambda_- = \lambda$, equation (23) becomes $(\mathcal{D} - \lambda)^2 w = 0$. The two linearly independent solutions are $E_{\mu,1}(\lambda z)$ (eigenfunction of $\mathcal{D}$ at λ) and a second solution found by reduction of order. Setting $w_2(z) = E_{\mu,1}(\lambda z) \cdot v(z)$ and substituting into $(\mathcal{D} - \lambda)^2 w_2 = 0$ gives $(\mathcal{D} - \lambda)[E_{\mu,1}(\lambda z) \mathcal{D}v + v \mathcal{D}E_{\mu,1}(\lambda z)] - \lambda E_{\mu,1}(\lambda z) v (\mathcal{D} - \lambda)$; using $(\mathcal{D} - \lambda)E_{\mu,1}(\lambda z) = 0$ this reduces to $E_{\mu,1}(\lambda z) \mathcal{D}^2 v + 2\mathcal{D}E_{\mu,1}(\lambda z) \mathcal{D}v = 0$, i.e. $\mathcal{D}^2 v + 2\lambda \mathcal{D}v = 0$, giving $\mathcal{D}v = -2\lambda v$ hence $v(z) = E_{\mu,1}(-2\lambda z)$; normalising shows the independent solution is $-\lambda z E_{\mu,1+\mu}(\lambda z)$. Direct verification: applying $\mathcal{D}$ to $z E_{\mu,1+\mu}(\lambda z) = \sum_{k=0}^{\infty} \lambda^k z^{k+1} / \Gamma(k\mu + \mu + 1)$ term by term,

$$\mathcal{D}[z^{k+1} / \Gamma(k\mu + \mu + 1)] = \frac{\Gamma((k+1)\mu + 1)}{\Gamma(k\mu + 1)} \cdot \frac{z^k}{\Gamma(k\mu + \mu + 1)} = \frac{z^k}{\Gamma(k\mu + 1)},$$

so $\mathcal{D}[z E_{\mu,1+\mu}(\lambda z)] = E_{\mu,1}(\lambda z)$. Applying $\mathcal{D}$ again: $\mathcal{D}[E_{\mu,1}(\lambda z)] = \lambda E_{\mu,1}(\lambda z)$. Thus $\mathcal{D}^2[z E_{\mu,1+\mu}(\lambda z)] = \lambda E_{\mu,1}(\lambda z)$ and $(\mathcal{D} - \lambda)^2[-\lambda z E_{\mu,1+\mu}(\lambda z)] = 0$, confirming (26) is indeed a solution. Imposing $w(0) = 1$ and $\mathcal{D}w(0) = \lambda - \lambda \cdot 1 = 0$ is satisfied by (26).

(iii) Each $E_{\mu,1}(\lambda z)$ is entire of order $1/\mu$ (Remark 3.3). The function $z E_{\mu,1+\mu}(\lambda z)$ is entire of the same order. A finite linear combination of entire functions of order $\leq 1/\mu$ is entire of order $\leq 1/\mu$; since the combination is non-trivial and each summand has exact order $1/\mu$, the order of w is exactly $1/\mu$. $\square$

3.4. Meromorphy of g on $\mathbb{C}$.

Theorem 3.7 (Meromorphy). *The function $g(z) = y(z^{1/\mu})$, initially defined for $|z| < R$, extends to a meromorphic function on all of $\mathbb{C}$. Its poles are precisely the zeros of w , form a discrete set with no finite accumulation point, and each pole is simple.*

Proof. From Theorem 3.5, $g = -\mathcal{D}w/(c_2w)$.

Both w and $\mathcal{D}w$ are entire. w is entire by Theorem 3.6. $\mathcal{D}w = -c_2gw$ (from (22)), and using the explicit form: in the non-degenerate case, $\mathcal{D}w = A\lambda_+E_{\mu,1}(\lambda_+z) + B\lambda_-E_{\mu,1}(\lambda_-z)$, a finite linear combination of entire functions, hence entire.

g is meromorphic. g is the quotient of two entire functions ($\mathcal{D}w$ and c_2w), hence meromorphic on $\mathbb{C}$ with poles at the zeros of w .

Discreteness. $w(0) = 1 \neq 0$, so $w \not\equiv 0$. By the identity theorem, the zeros of w are isolated and have no finite accumulation point.

Every pole of g is simple. Suppose $w(z_0) = 0$ and $\mathcal{D}w(z_0) = 0$ simultaneously. Then from (23), $\mathcal{D}^2w(z_0) = c_1\mathcal{D}w(z_0) - c_0c_2w(z_0) = 0$. Applying $\mathcal{D}$ repeatedly to (23) shows all higher $\mathcal{D}^k w(z_0) = 0$, so the power series of w at z_0 vanishes identically, contradicting $w \not\equiv 0$. Hence $\mathcal{D}w(z_0) \neq 0$ at every zero of w , and

$$\text{Res}_{z=z_0} g(z) = \lim_{z \rightarrow z_0} (z - z_0) \cdot \frac{-\mathcal{D}w(z)}{c_2w(z)} = \frac{-\mathcal{D}w(z_0)}{c_2w'(z_0)},$$

which is finite and nonzero, confirming a simple pole. $\square$

Corollary 3.8 (Unconditional global convergence). *Under the assumption that y is bounded on $[0, \infty)$ (equivalently, g has no poles on the positive real z -axis), the following hold without any further hypothesis on g :*

- (i) Nuttall–Pommerenke (Theorem 4.5) applies: $R_M \rightarrow g$ in capacity on $\mathbb{C} \setminus E$, $\text{cap}(E) = 0$.
- (ii) De Montessus de Ballore (Theorem 4.4) applies on every disc $|z| < \rho$ avoiding zeros of w : $R_M \rightarrow g$ uniformly on compact subsets.
- (iii) $y(t) = \lim_{M \rightarrow \infty} R_M(t^\mu)$ pointwise for every $t \geq 0$.

The Puiseux radius equals $R = \min\{|z| : w(z) = 0\}$, computable from the Mittag-Leffler combination in Theorem 3.6.

4. PADÉ RESUMMATION IN $z = t^\mu$

4.1. Power series in z and the Hankel system. Recall $g(z) = \sum_{k=1}^{\infty} a_k z^k$ for $|z| < R$. Fix $M \in \mathbb{N}$ and seek polynomials

$$P_M(z) := \sum_{k=0}^M p_k z^k, \quad Q_M(z) := 1 + \sum_{k=1}^M q_k z^k$$

satisfying

$$(27) \quad Q_M(z)g(z) - P_M(z) = \mathcal{O}(z^{2M+1}) \quad (z \rightarrow 0).$$

Lemma 4.1 (Hankel form of the denominator system). *Matching coefficients of z^n in (27) gives $p_0 = 0$,*

$$(28) \quad p_n = a_n + \sum_{j=1}^{\min(n,M)} q_j a_{n-j} \quad \forall 1 \leq n \leq M,$$

$$(29) \quad 0 = a_n + \sum_{j=1}^M q_j a_{n-j} \quad \forall M+1 \leq n \leq 2M.$$

The denominator system (29) is the Hankel linear system $H_M \mathbf{q} = -\mathbf{b}$,

$$(30) \quad H_M = \begin{pmatrix} a_M & a_{M-1} & \cdots & a_1 \\ a_{M+1} & a_M & \cdots & a_2 \\ \vdots & \vdots & \ddots & \vdots \\ a_{2M-1} & a_{2M-2} & \cdots & a_M \end{pmatrix}, \quad \mathbf{q} = \begin{pmatrix} q_1 \\ \vdots \\ q_M \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} a_{M+1} \\ \vdots \\ a_{2M} \end{pmatrix}.$$

Proof. Expand $Q_M(z)g(z) = (1 + \sum_{j=1}^M q_j z^j) \sum_{k \geq 1} a_k z^k$. The coefficient of z^n is $a_n + \sum_{j=1}^{\min(n, M)} q_j a_{n-j}$ (with $a_i = 0$ for $i \leq 0$). Equating to the coefficient of z^n in $P_M(z)$: for $1 \leq n \leq M$ this gives (28), and for $M+1 \leq n \leq 2M$ (where P_M contributes zero) gives (29). Writing (29) in matrix form yields (30). $\square$

Remark 4.2 (Invertibility). H_M is invertible iff g is not a rational function of degree $< M$. For $\mu \in (0, 1)$, $g = -\mathcal{D}w/(c_2 w)$ has infinitely many poles (the zeros of the entire function w), so g is not rational and $\det H_M \neq 0$ for all $M \geq 1$ generically.

The diagonal Padé approximant is

$$(31) \quad R_M(z) := \frac{P_M(z)}{Q_M(z)} = \frac{\sum_{k=0}^M p_k z^k}{1 + \sum_{k=1}^M q_k z^k},$$

defined on $\mathcal{D}_M := \{z \in \mathbb{C} : \det H_M \neq 0, Q_M(z) \neq 0\}$.

4.2. A-posteriori error bound. Define $\Delta_k(z) := R_k(z) - R_{k-1}(z)$ for $k \geq 1$.

Assumption 1 (Uniform geometric tail). At a fixed $z \in \mathcal{D}_M \cap \mathcal{D}_{M-1}$:

- (a) g has no poles in $\{|\zeta| \leq |z|\}$;
- (b) $Q_M(z) \neq 0$ and $Q_{M-1}(z) \neq 0$;
- (c) there exists $\rho \in (0, 1)$ such that $|\Delta_{k+1}(z)| \leq \rho |\Delta_k(z)|$ for every $k \geq M-1$.

Theorem 4.3 (Computable error bound). *Under Assumption 1,*

$$(32) \quad |g(z) - R_M(z)| \leq \frac{\rho |\Delta_M(z)|}{1 - \rho}.$$

Taking the observable ratio $\rho_* := |\Delta_M(z)|/|\Delta_{M-1}(z)|$,

$$(33) \quad |g(z) - R_M(z)| \leq \frac{|\Delta_M(z)|^2}{|\Delta_{M-1}(z)| - |\Delta_M(z)|}.$$

Proof. By Assumption 1(c), $|\Delta_{M+j}(z)| \leq \rho^j |\Delta_M(z)|$ for all $j \geq 0$ (induction: base $j = 0$ trivial; step from $|\Delta_{M+j+1}| \leq \rho |\Delta_{M+j}| \leq \rho^{j+1} |\Delta_M|$). The telescoping series $g(z) - R_M(z) = \sum_{k=M+1}^{\infty} \Delta_k(z)$ converges absolutely, and

$$|g(z) - R_M(z)| \leq \sum_{j=1}^{\infty} |\Delta_{M+j}(z)| \leq |\Delta_M(z)| \sum_{j=1}^{\infty} \rho^j = \frac{\rho |\Delta_M(z)|}{1 - \rho},$$

giving (32). Substituting $\rho = \rho_* = |\Delta_M|/|\Delta_{M-1}|$ yields (33). $\square$

4.3. Global convergence on the positive real axis.

Theorem 4.4 (de Montessus de Ballore). *Let g be meromorphic in $\{|z| < \rho\}$ with exactly n poles (counted with multiplicity) and no poles on $|z| = \rho$. The diagonal Padé approximants R_M converge to g uniformly on compact subsets of $\{|z| < \rho\} \setminus \{\text{poles}\}$ as $M \rightarrow \infty$.*

Theorem 4.5 (Nuttall–Pommerenke). *Let g be meromorphic in $\mathbb{C}$ with at most countably many poles and no finite accumulation point. The diagonal sequence R_M converges to g in capacity on $\mathbb{C} \setminus E$, $\text{cap}(E) = 0$.*

Theorem 4.6 (Global Padé–Müntz representation). *Let $y(t)$ solve (10)–(11) and be bounded on $[0, \infty)$. Then*

$$(34) \quad y(t) = \lim_{M \rightarrow \infty} R_M(t^\mu) = \lim_{M \rightarrow \infty} \frac{P_M(t^\mu)}{Q_M(t^\mu)} \quad \forall t \geq 0.$$

Proof. Meromorphy of g on $\mathbb{C}$ is Theorem 3.7 (unconditional). Its poles are discrete with no finite accumulation point, so Theorem 4.5 applies: $R_M \rightarrow g$ in capacity on $\mathbb{C} \setminus E$. Since g is holomorphic on a neighbourhood of every point of $(0, \infty)$ (boundedness of y excludes positive real poles), Theorem 4.4 gives uniform convergence on compact subsets of $(0, \infty)$, hence pointwise at every $z = t^\mu > 0$. At $t = 0$: $y(0) = 0$ and $R_M(0) = p_0/1 = 0$ for every M . $\square$

4.4. Algorithmic summary. Given (c_0, c_1, c_2, μ) and Padé order M :

- (1) Compute $a_1, \dots, a_{2M}$ by (14)–(15). Cost: $\mathcal{O}(M^2)$.
- (2) Form H_M and $\mathbf{b}$ from (30).
- (3) Solve $H_M \mathbf{q} = -\mathbf{b}$. Cost: $\mathcal{O}(M^3)$ (dense LU); reducible to $\mathcal{O}(M^2)$ via the Levinson–Durbin algorithm.
- (4) Recover numerator via (28).
- (5) Evaluate $y_M(t) = P_M(t^\mu)/Q_M(t^\mu)$ at any $t \geq 0$. Cost: $\mathcal{O}(M)$ per query.

Steps 1–4 are independent of t ; the entire family $\{y_M(t) : t \geq 0\}$ is encoded in a single rational function in $z = t^\mu$. There is no grid, no time stepping, no Newton iteration, no fixed-point loop.

APPENDIX A. MITTAG-LEFFLER FUNCTIONS: ORDER AND TYPE

Proposition A.1. *For $\mu \in (0, 1)$ and $\lambda \in \mathbb{C}$, $z \mapsto E_{\mu,1}(\lambda z)$ is entire of order $\rho = 1/\mu$ and type $\sigma = |\lambda|^{1/\mu}$.*

Proof. The Taylor coefficients are $c_k = \lambda^k / \Gamma(k\mu + 1)$. By Stirling’s formula, $\log \Gamma(k\mu + 1) = k\mu \log(k\mu) - k\mu + \frac{1}{2} \log(2\pi k\mu) + \mathcal{O}(1/k)$, so $\log |c_k|^{-1/k} \sim \mu \log(k\mu) - \mu - \log |\lambda| \rightarrow \infty$, confirming infinite radius of convergence. The Hadamard order formula gives

$$\rho = \limsup_{k \rightarrow \infty} \frac{k \log k}{\log(1/|c_k|)} = \limsup_{k \rightarrow \infty} \frac{k \log k}{k\mu \log k - k\mu \log \mu + k\mu + \mathcal{O}(\log k)} = \frac{1}{\mu}.$$

The type at order $1/\mu$ follows from $\sigma = (e\rho)^{-1} \limsup_{k \rightarrow \infty} k^{1/\mu} |c_k|^{\mu/k}$, which evaluates to $|\lambda|^{1/\mu}$. $\square$

APPENDIX B. CLOSED-FORM SPECIAL CASES

Case $c_1 = 0$. $\lambda_\pm = \pm \sqrt{-c_0 c_2}$, $A = B = 1/2$, and

$$w(z) = \frac{1}{2} [E_{\mu,1}(\sqrt{-c_0 c_2} z) + E_{\mu,1}(-\sqrt{-c_0 c_2} z)] = E_{2\mu,1}(-c_0 c_2 z^2),$$

using the standard reflection identity $\frac{1}{2} [E_{\mu,1}(u) + E_{\mu,1}(-u)] = E_{2\mu,1}(-u^2)$. The poles of g are at the zeros of $E_{2\mu,1}(-c_0 c_2 z^2)$.

Case $c_0 = 0$. $\lambda_+ = c_1$, $\lambda_- = 0$. The formula (25) gives $A = 1$, $B = 0$, so $w(z) = E_{\mu,1}(c_1 z)$ and

$$g(z) = -\frac{c_1 E_{\mu,1}(c_1 z)}{c_2 E_{\mu,1}(c_1 z)} = -\frac{c_1}{c_2},$$

the constant fixed point, consistent with $D^\mu(-c_1/c_2) = 0 = c_1(-c_1/c_2) + c_2(-c_1/c_2)^2$.

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